```php
<?php
    /*  3 b. Develop a PHP Program to compute the roots of a quadratic
            equation by accepting the coefficients. Print the
            appropriate messages.

        3b_roots.php
    */
    /*  Generic quadratic equation, second-degree polynomial equation
                                        2
                          f(x) = y = a x  + b x + c = 0
        where a and b known as coefficients, and c , all are constants
            a ≠ 0, else it would be a linear equation, bx + c

        Can have up to (atmost) two solutions, known as the roots
            values of x, where equation evaluates to zero
            graphically, where curve intersects with x axis, y = 0
                Example, on x y cartesian coordinate system
                    x*x - 5 = 0, as ax2 + bx + c = 0, a = 1, b = 0, c = -5

                                     y = f(x) = x*x - 5
                                     ↑
                                     |
                             \      3|       /
                              \     2|      /
                    _ _ _ _ _  @ _ _ 1._ _ @ _ _ _ _ _ _→ x
                        -5  -3 \  -1 | 1  / 3   5   7
                               \   -1|    /
            Root 2 (-2.2, 0) @  \  -2|   /   @ Root 1 (2.2, 0)
                                \  -3|   /
                                 \ -4|  /
                                   \ - /
                                     |        . (0, 0) origin

                    Visualize curve,roots on GeoGebra, Graphing Calculator
                        https://www.geogebra.org/graphing
    */
    /*                         2
        Roots = (  -b  ±  √ ( b - 4 a c )  ) / 2 a                Why/How?

        √ ( b*b - 4*a*c ) , decides if roots are real (ℝ) or complex (C)

        b*b - 4*a*c, also known as discriminant Δ
    */
    /*                            2
        numerator   = -b  ±  √ ( b - 4 a c )
        denominator = 2 a

        Roots = numerator / denominator
    */
?>
```

```
<?php

    $a = 1;
    $b = 1; // check with values of b as 1 , or 2 , or 3
    $c = 1;

    echo "<br /> For a = $a , b = $b , c = $c , roots are : ";
                                                    //     2
    $disc = $b * $b - 4 * $a * $c;  // Δ , discriminant , b - 4ac
    $denom = 2 * $a ;

    if ( $disc == 0 )                                    // √ 0 is real
     {  // since Δ $disc is 0 , roots = ( -b ± √ 0 ) / 2 a = -b/2a
        $root = -$b / $denom;
        echo "<br /> real , same root = " . $root ;
     }
    if ( $disc > 0 ) // else if can also be used, in PHP keyword is elseif
     {  // √ Δ is real , and two possible values , hence two real roots
        $root1 = ( -$b + sqrt($disc) ) / $denom; //( -b + √ (bb - 4ac) )/2a
        $root2 = ( -$b - sqrt($disc) ) / $denom; //( -b - √ (bb - 4ac) )/2a
        echo "<br /> real, two distinct roots, r1 = $root1 , r2 = $root2";
     }
    if ( $disc < 0 )                    // if elseif else can also be used
     {  // Δ is -ve , √ -ve value is complex, represented using i ,i = √ -1
        //  root has to broken down as real_part + i imaginary_part
        $real_part = -$b / $denom;
        $imag_part = sqrt( - $disc ) / $denom;  //√ double negation, √ -(Δ)
        echo "<br /> complex, two distinct roots <br /> ";
        echo "$real_part + i $imag_part, and $real_part - i $imag_part";
     }  // How could complex roots be displayed graphically
?>

<!--
For more on , please check ,
    sqrt — Square root                                      $ php --rf sqrt
        https://www.php.net/manual/en/function.sqrt.php

instead of square root function, power function can also be used
    pow — Exponential expression                             $ php --rf pow
        https://www.php.net/manual/en/function.pow.php
-->
```